

AI Quality Incident Assistant

An AI Strategy Proposal for
Hertzmänn & Söhne



HERTZMANN
& SÖHNE

PRECISION. QUALITY. RELIABILITY.

About Hertzmann & Söhne

Company Overview



German automotive manufacturer



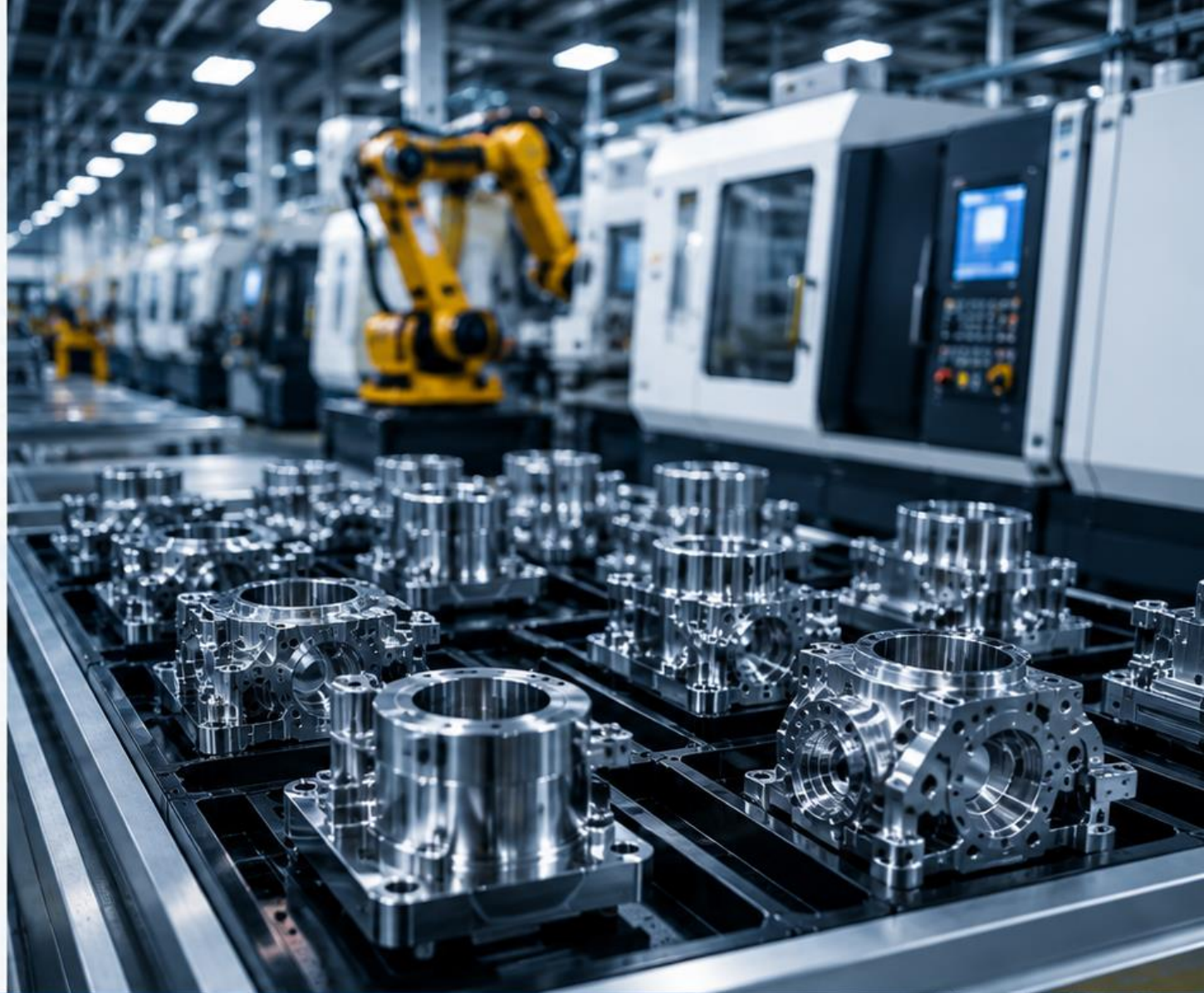
Produces high-precision
automotive components



Operates under strict
quality standards



Strong focus on product
quality and traceability



Quality and compliance are critical throughout the manufacturing process.

Current Challenge

The current process is manual, time-consuming, and prone to inconsistencies.



Challenges

-  Time-consuming documentation
-  Repetitive administrative work
-  Inconsistent report structure
-  Missing information requiring revision

Can AI help?

Can generative AI reduce documentation effort while keeping engineers fully responsible for quality decisions?

Current



Manual extraction



Manual formatting



Manual completeness check



Proposed



AI extracts facts



AI drafts report



Engineer validates



Proposed Solution

An AI-powered assistant that streamlines quality documentation while keeping engineers in full control.



What this solution delivers



Less documentation effort
Reduces time spent on repetitive administrative tasks



More consistent reports
Structured output improves consistency and completeness



Fewer missing details
AI helps identify missing information early



Full engineer accountability
Engineers remain in control of every decision



AI supports documentation. Engineers remain responsible for all technical decisions.

Why Generative AI?

Traditional Rule-Based Automation



- ✗ Requires predefined templates
- ✗ Difficult to process free-text engineer notes
- ✗ Limited flexibility
- ✗ High maintenance effort

VS.

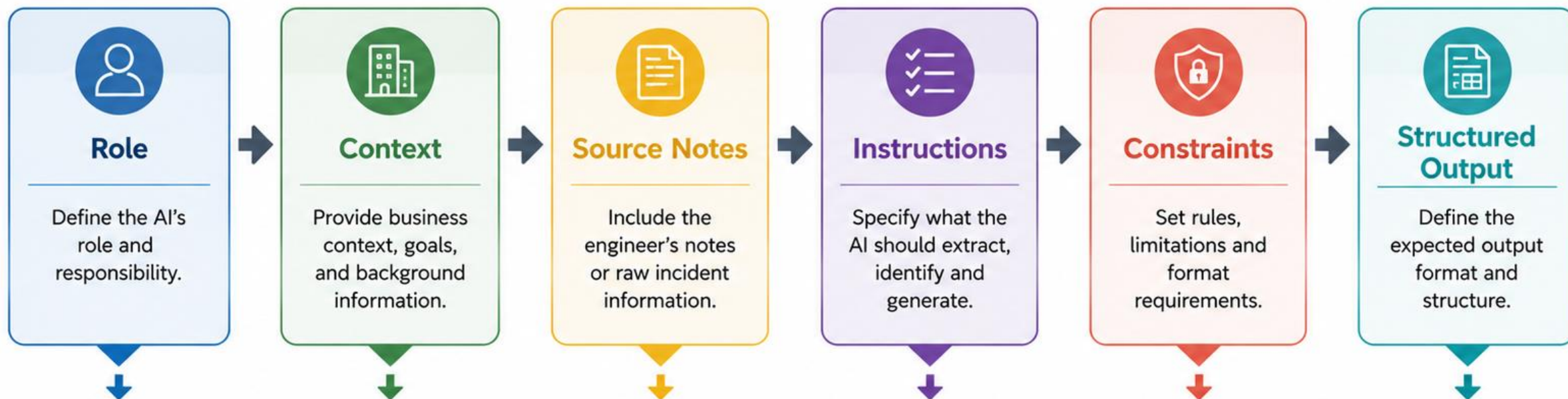
Generative AI (LLM)

- ✓ Understands natural language
- ✓ Extracts relevant information
- ✓ Organizes data into a structured report
- ✓ Adapts to different writing styles



Generative AI is ideal because quality incident notes are written in natural language rather than fixed templates.

Prompt Engineering



Why this structure?



Defines the AI's responsibility

Clearly sets the role and boundaries of the assistant.



Reduces ambiguity

Provides all necessary information in a clear and organized way.



Prevents hallucinations

Constraints and rules keep the AI grounded in the provided information.



Produces consistent reports

A structured approach leads to reliable and repeatable outputs.



The prompt was refined after testing to improve output consistency.

Prompt Validation

Three test cases were designed to evaluate accuracy, completeness, and consistency of the prompt.



Test Case 1 Dimensional Deviation



Purpose

Verify preservation of measurements.



Result

 **Passed**



Test Case 2 Documentation Issue



Purpose

Verify recognition of missing documentation.



Result

 **Passed**



Test Case 3 Surface Defect



Purpose

Verify classification of cosmetic defects.



Result

 **Passed**



INPUT – Engineer Notes (Example)

Part: Bracket A123
Date: 2025-05-12
Inspector: J. Smith
Notes:
Measured hole diameter $\varnothing 10.35$ mm.
Drawing specifies $\varnothing 10.00 \pm 0.10$ mm.
Out of tolerance.
Lot 25-0417.
Cause unknown.
Photos attached.



10:34 AM



OUTPUT – Claude Response (Structured NCR Draft)

Part / Item	Bracket A123
Issue Description	Hole diameter out of tolerance.
Measurement	Measured: $\varnothing 10.35$ mm Specified: $\varnothing 10.00 \pm 0.10$ mm
Deviation	+0.25 mm (Out of tolerance)
Lot / Serial	Lot 25-0417
Probable Cause	Unknown
Attachments	Photos attached
Notes	NCR draft – engineer review required.

Classification

Dimensional Deviation

Missing Information

- Root cause
- Proposed corrective action

Status

Draft – Pending Engineer Review

10:34 AM



Business Value



Improved Efficiency

Less time spent preparing reports.



Better Documentation

More consistent and complete NCRs.



More Engineering Focus

More time for investigations and process improvement.

Estimated Business Impact

Current effort

2–3

hours/week



AI drafting reduction

≈40%



Estimated monthly saving

3–5

hours
per engineer



Pilot assumptions — actual benefits should be validated during implementation.

Recommendations & Closing

Start with a controlled pilot, measure results, and then decide on wider implementation.

1



Pilot

- Deploy the AI Quality Incident Assistant
- Use with a small group
- Collect feedback



2



Measure

- Documentation time
- Report consistency
- User satisfaction



3



Evaluate

- Assess impact
- Refine the solution
- Decide on wider implementation



**AI should support quality engineers,
not replace them.**

This solution reduces administrative work, improves documentation quality, and keeps engineers focused on investigations and continuous improvement.